



Mathematics
Pathway 2
Test

Baldivis
Secondary College

Name: Mark Inkey Class: _____ Score: ____/57
Resources Allowed: Calculator, 1 page of notes (A4 one sided), Formula sheet

Time: 55mins

Marks: 57

1. [4 marks]

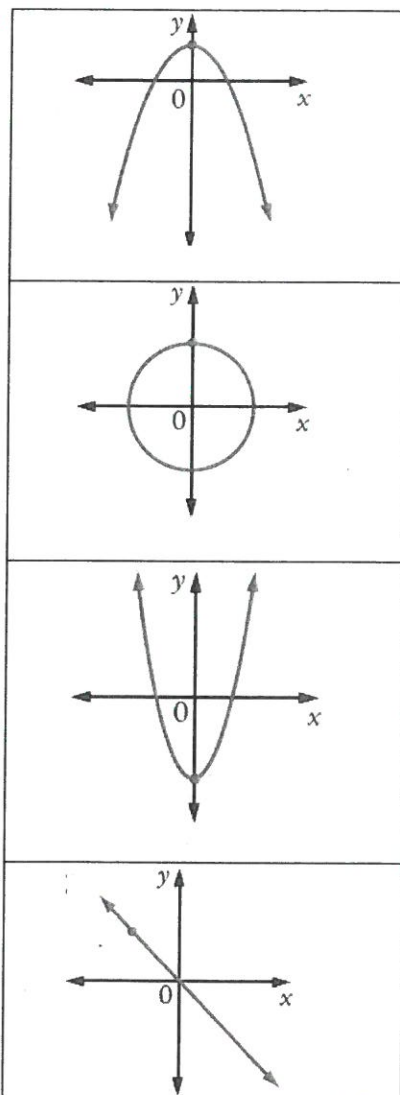
Match the following equations to the appropriate graph (circle, parabolas, linear)

$$y = -2x$$

$$y = (x + 3)(x - 3)$$

$$y = -2x^2 + 3$$

$$y^2 + x^2 = 3$$



2. [3,3 = 6 marks]

Solve the following simultaneously

a) $x + y = 5$
 $x = y - 1$

$$\begin{aligned} y-1+y &= 5 \\ 2y-1 &= 5 \\ 2y &= 6 \\ y &= 3 \end{aligned}$$

$$\begin{aligned} x &= 3-1 \\ x &= 2 \end{aligned}$$

$$x=2, y=3.$$

b)

$$7x + y = 128$$

$$+ 4x - y = 15$$

$$11x = 143$$

$$x = 13$$

$$4(13) - y = 15$$

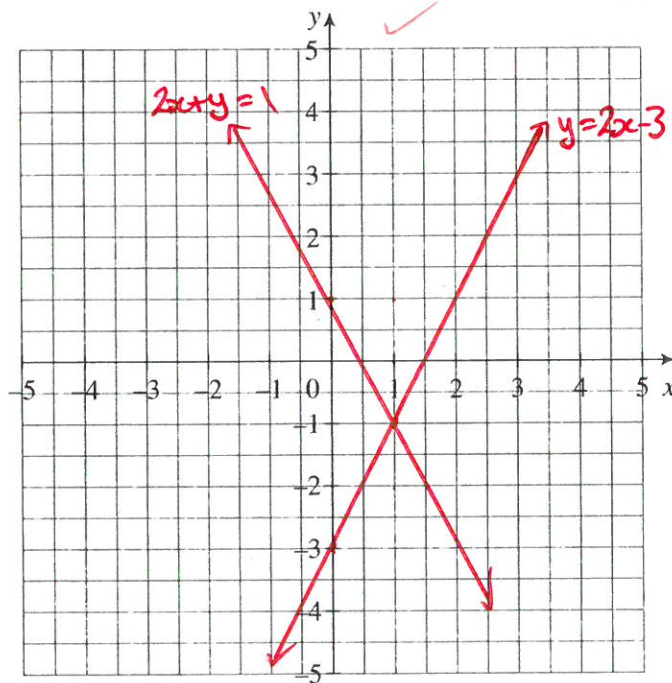
$$y = 37$$

$$x=13, y=37$$

3. [2,1= 3marks]

a) Sketch the graphs with equations

$y = 2x - 3$ and $2x + y = 1$ accurately on the set of axes provided.



b) Solve the equations graphically by stating the x and y values at the point of intersection.

$$(1, -1).$$

$$x=1$$

$$y=-1$$

4. [2, 3 = 5 marks]

Entry to an entertainment park costs \$150 for one adult and three children. The price of an adult ticket is \$30 more than the child's ticket.

a) Write two equations to represent this situation

$$a + 3c = 150 \quad \checkmark$$

$$a = c + 30 \quad \checkmark$$

b) Using the two equations from a) solve the equations simultaneously and hence determine the price of an adult ticket and a child ticket.

$$c + 30 + 3c = 150$$

$$4c + 30 = 150 \quad \checkmark$$

$$4c = 120$$

$$c = 30 \quad \checkmark$$

$$a = 30 + 30$$

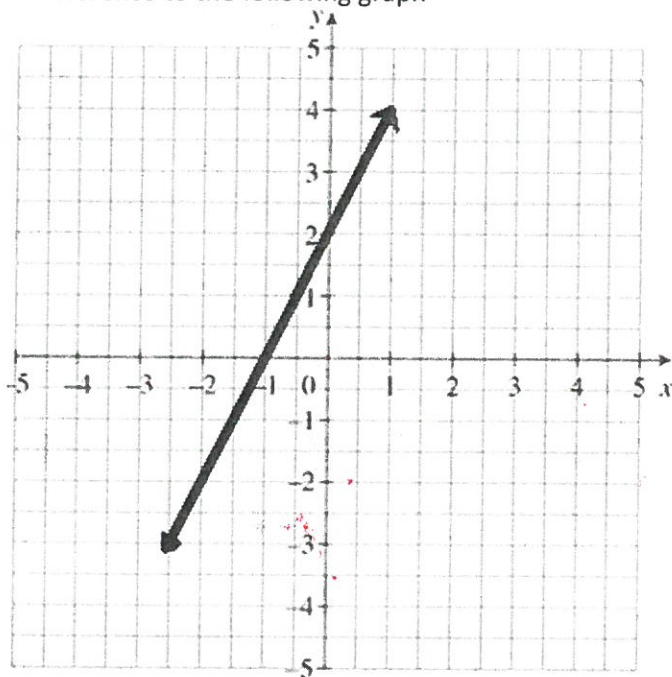
$$a = 60 \quad \checkmark$$

Adult Ticket = \$60

Child Ticket = \$30

5. [3, 2, 2 = 7 marks]

In reference to the following graph



a) State the;

i) gradient

$$2 \quad \checkmark$$

ii) y-intercept

$$(0, 2) \quad \checkmark$$

iii) x-intercept

$$(-1, 0) \quad \checkmark$$

b) Write the equation for the line

$$y = 2x + 2 \quad \checkmark$$

c) Write an equation for a line that is

i) Perpendicular

$$y = -\frac{1}{2}x + \quad \checkmark$$

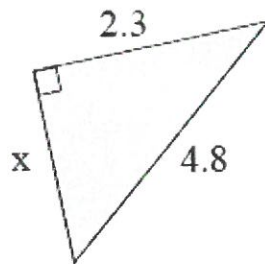
ii) Parallel

$$y = 2x + \quad \checkmark$$

6. [2,2 = 4 marks]

Determine the value of missing side using Pythagoras

a)



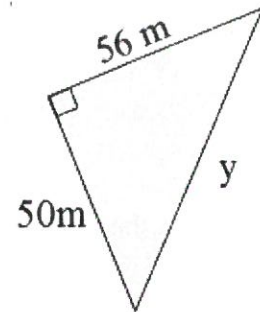
$$b = \sqrt{c^2 - a^2}$$

$$b = \sqrt{4.8^2 - 2.3^2}$$

$$b = 4.213 \text{ (3 d.p.)}$$

$$x = 4.2$$

b)



$$c = \sqrt{a^2 + b^2}$$

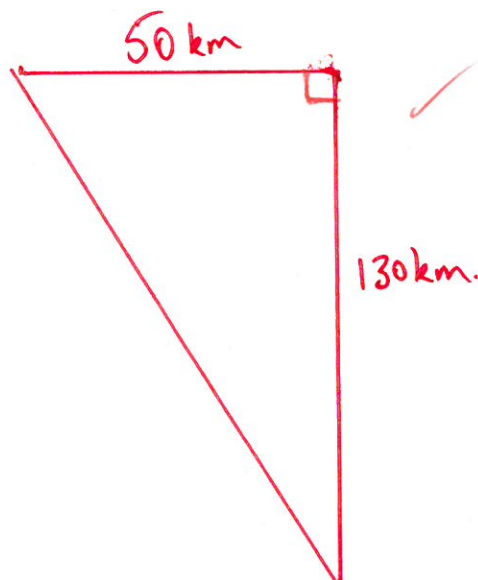
$$y = \sqrt{56^2 - 50^2}$$

$$y = 75.07 \text{ (2 d.p.)}$$

$$y = 75 \text{ m.}$$

7. [3 marks]

A plane flies 50km due East and then 130km due South. How far is the plane from its starting point? (hint: draw a diagram)



$$c = \sqrt{a^2 + b^2}$$

$$c = \sqrt{50^2 + 130^2}$$

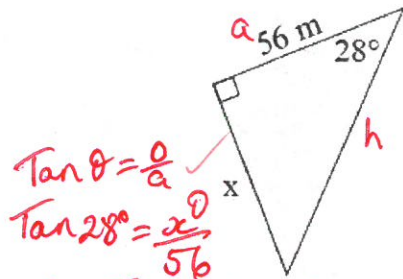
$$c = 139.28 \text{ (2 d.p.)}$$

The plane is 139.28 km from its starting point.

8. [2, 2, 2 = 6 marks]

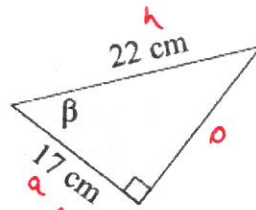
Use the trigonometric ratios to determine the value of the pronumerals in the following diagrams.

a)



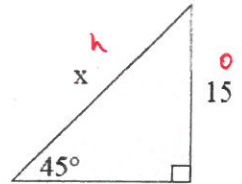
$$\begin{aligned}\tan \theta &= \frac{o}{a} \\ \tan 28^\circ &= \frac{x}{56} \\ x &= \tan 28^\circ \times 56 \\ x &= 29.78 \text{ m (2dp)}\end{aligned}$$

b)



$$\begin{aligned}\cos \beta &= \frac{a}{h} \\ \cos \beta &= \frac{17}{22} \\ \cos^{-1} \beta &= 39.38 \text{ (2dp)} \\ \beta &= 39^\circ\end{aligned}$$

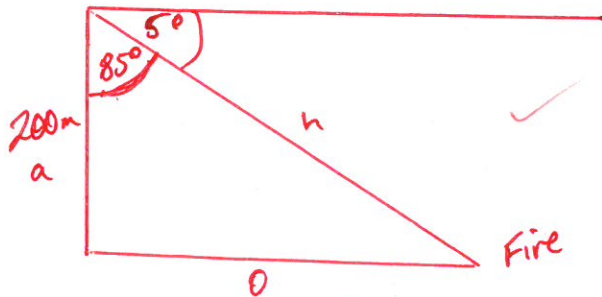
c)



$$\begin{aligned}\sin \theta &= \frac{o}{h} \\ \sin 45^\circ &= \frac{15}{x} \\ x &= \frac{15}{\sin 45^\circ} \\ x &= 21.21 \text{ (2dp)} \\ x &= 21\end{aligned}$$

9. [3 marks]

From the top of a 200m tower, the angle of depression to a fire is 5 degrees. How far away from the base of the tower is the fire?



$$\begin{aligned}\tan \theta &= \frac{o}{a} \\ \tan 85^\circ &= \frac{\text{opp}}{200} = \frac{x}{200} \\ x &= \tan 85^\circ \times 200 \\ x &= 2286.01\end{aligned}$$

The fire is 2,286.01m away from the base of the tower.

10. [2,3,3 = 8 marks]

Solve the following Quadratics

a) $(x + 4)(x - 2) = 0$

$$\begin{aligned}x + 4 &= 0 \text{ or } x - 2 = 0 \\ x &= -4 \text{ or } x = 2\end{aligned}$$

b) $x^2 - 2x - 15 = 0$

$$\begin{aligned}(x - 5)(x + 3) &= 0 \\ x - 5 &= 0 \text{ or } x + 3 = 0 \\ x &= 5 \text{ or } x = -3\end{aligned}$$

c) $x^2 + 5x + 6 = 0$

$$\begin{aligned}(x + 3)(x + 2) &= 0 \\ x + 3 &= 0 \text{ or } x + 2 = 0 \\ x &= -3 \text{ or } x = -2\end{aligned}$$

11. [1,2,2,2 = 7 marks]

For the quadratic equation

$$y = x^2 + 4x + 3$$

a) State the y-intercept

make $x=0$.
y-intercept = 3. ✓

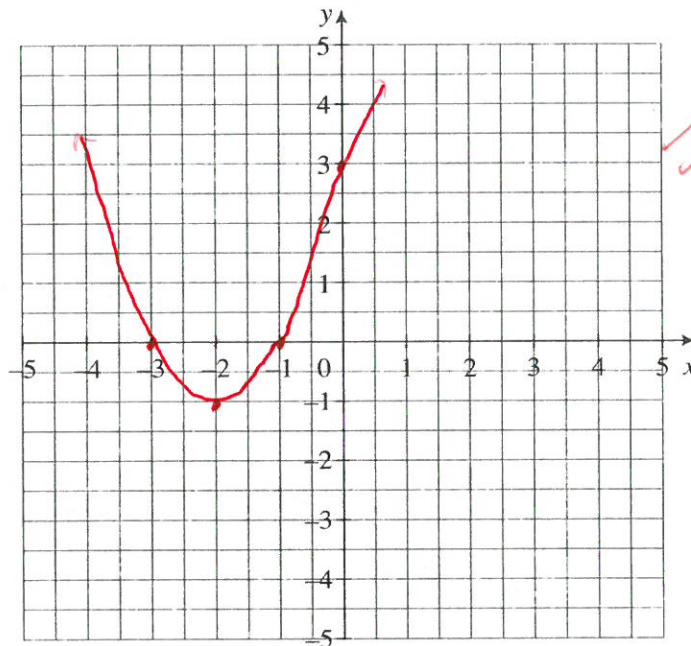
b) The x-intercepts

make $y=0$.
 $0 = x^2 + 4x + 3$.
 $0 = (x+3)(x+1)$.
 $x = -3$ or $x = -1$.
x intercepts = -3 and -1. ✓

c) The turning point

$x = \frac{-b}{2a} = \frac{-4}{2} = -2$. ✓
 $y = -2^2 + 4(-2) + 3$.
 $y = -1$. ✓
Turning Point $(-2, -1)$. ✓

d) Sketch the graph on the



End of Test